

Scleritis is not commonly encountered in Vogt-Koyanagi-Harada syndrome since the disease is caused by an autoimmune attack directed against melanin-bearing cells. Meningeal signs and symptoms represent an inflammatory process in the meninges because of the presence of residual melanocytes derived from the neural crest. The sclera may contain melanocytes, especially at the site of exit of the long ciliary nerves anterior to the recti muscles. The autoimmune attack on melanocytes in the sclera may lead to focal scleritis and may cause the destruction of the collagen fibers.

References

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Safety and Effectiveness of Magnetic Resonance Imaging of Choroidal Melanoma Patients With Episcleral Tantalum Rings After Proton Beam Irradiation

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During the last 12 years approximately 1,200 patients with choroidal melanomas have been treated with proton beam irradiation at the Harvard Cyclotron. Before irradiation treatment, the patients are admitted to the Massachusetts Eye and Ear Infirmary for the suturing of four 2.5-mm diameter tantalum rings, which serve as reference markers for radiographic alignment of the tumor with the proton beam.¹ These rings remain permanently sutured on the sclera.

Recently, concern has been expressed as to whether magnetic resonance imaging can be performed safely and effectively after proton beam irradiation on choroidal melanoma patients with episcleral tantalum rings. Magnetic resonance imaging is now being used to assess the location and size of choroidal melanomas, and is likely to be used more in the future. Magnetic resonance is known to be a safe, noninvasive technique with many medical applications; potential hazards have so far only been known to exist for the imaging of subjects with ferromagnetic foreign bodies or cardiac pacemakers.^{2,3} Hazards associated with tantalum rings may include spatial disruption of the rings caused by magnetic torque or heating of the rings induced by radiofrequency pulses. Effective magnetic resonance imaging depends on the absence of artifacts resulting from the rings. Such artifacts make the evaluation of these eyes with computed tomography difficult.

Pure tantalum is nonmagnetic and has been shown to have no torque in a clinical magnetic resonance imaging magnet.⁴ We attempted to determine whether tantalum might experience an increase in temperature as the result of a series of radiofrequency pulses. A temperature change of a few degrees in the tantalum rings caused by the imaging process could cause injury to the eye. By using enucleated human eyes as subjects we also wanted to ascertain whether high quality images can be produced despite the presence of tantalum rings.

The temperature response of a tantalum ring to radiofrequency pulses was monitored by a microthermistor bead with a temperature coefficient of -3.9% C. The leads to the thermistor were tinned and soldered to two twisted insulated copper wires (26 G, 0.5 m long), which in turn were soldered to 10 m of stranded wire cable. The cable ends were connected to the junctions of a Wheatstone bridge with an elec-